TEST REPORT

Report No. : **ZKT-2603313487R**

Issue Date : Apr. 07, 2026

Applicant name : **Guangzhou Xingyi Electronic Technology Co., Ltd.**

Address : 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.

Sample Description

Sample name : ATK-CLIMX91S Core Board

Trade mark : ALIENTEK

Model No. : ATK-CLIMX91S

Manufacturer : Guangzhou Xingyi Electronic Technology Co., Ltd.

Address : 4th & 16th Floors, Building 2, Yudong Sci-Tech Park, No.3 Ketai 1st Road, Baiyun District, Guangzhou.

Test Requested

No.	Requested	Conclusion
1	EU RoHS Directive (EU)2015/863 Amending Annex II to Directive 2011/65/EU —Lead, Cadmium, Mercury, Hexavalent Chromium, PBBs and PBDEs Content —Di-(2-ethylhexyl) phthalate(DEHP), Benzylbutyl phthalate(BBP), Dibutyl phthalate (DBP), Diisobutyl phthalate(DIBP) Content	PASS

Reviewer by : _____ Approved by : _____

Shenzhen ZKT Technology Co., Ltd.
Approved

Testing Laboratory	:	**Shenzhen ZKT Technology Co., Ltd.**
Address	:	1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, ShenZhen, China.
Test Report Form No.	:	TRF-CL-001_V0
Received Date	:	Mar. 31, 2026
Testing Period	:	Mar. 31, 2026 to Apr. 07, 2026

Version

Report No.	Issue Date	Description	Approved
ZKT-2603313487R	Apr. 07, 2026	Original	Valid

Test Method:

1 XRF Screening

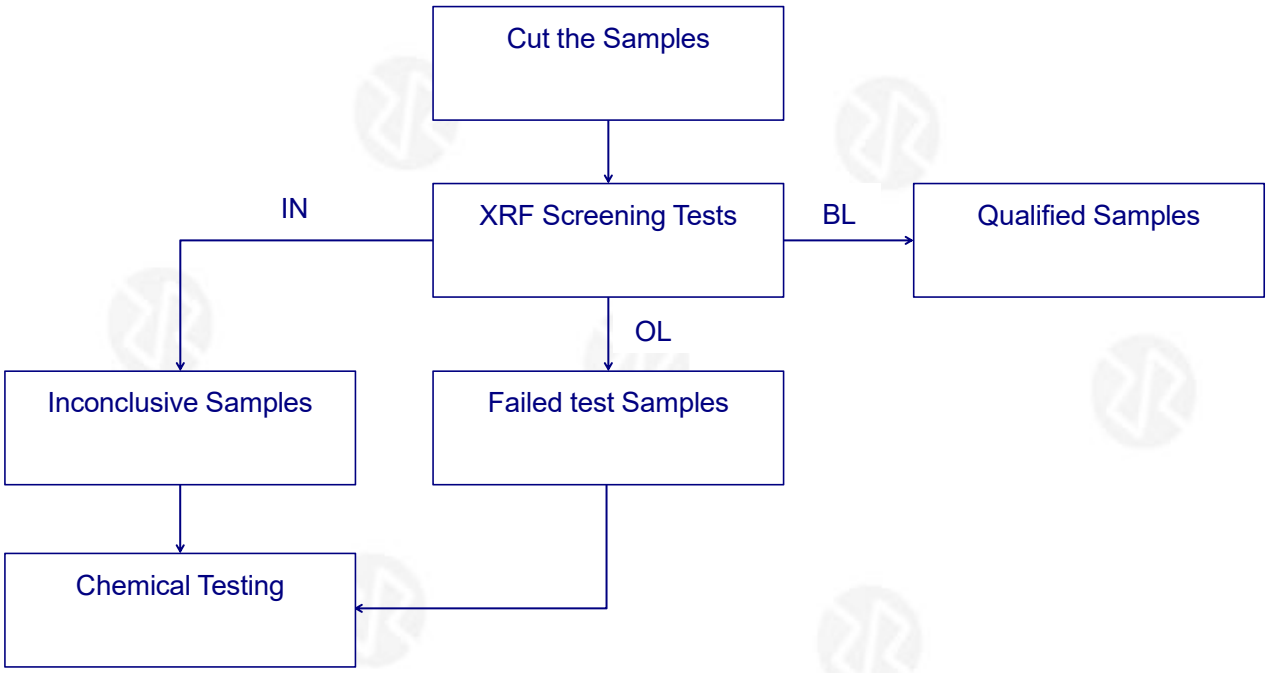
Test item(s)	Test Method(s)	Equipment(s)
Lead (Pb)	IEC 62321-3-1:2013	EDX LE
Cadmium (Cd)		
Mercury (Hg)		
Chromium (Cr)		
Bromine (Br)		

2 Chemical Testing

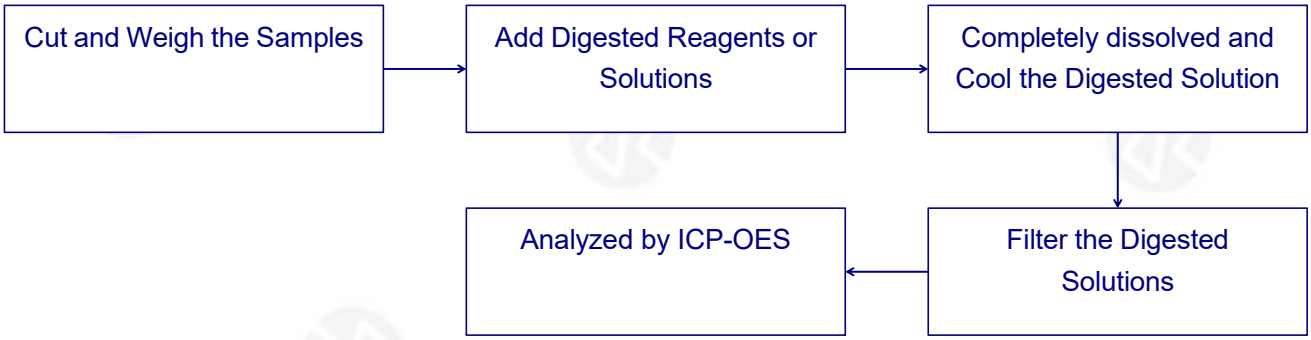
Test item(s)	Test Method(s)	Equipment(s)
Mercury(Hg)	IEC 62321-4:2013+A1:2017	ICP-OES
Lead(Pb)	IEC 62321-5:2013	ICP-OES
Cadmium(Cd)	IEC 62321-5:2013	ICP-OES
PBB	IEC 62321-6:2015	GC-MS
PBDE		
Chromium(Cr VI) for colourless and coloured corrosion-protected coatings on metals	IEC 62321-7-1:2015	UV-Vis
Chromium(Cr VI) for polymers and electronics	IEC 62321-7-2:2017	UV-Vis
DEHP	IEC 62321-8:2017	GC-MS
DBP		
BBP		
DIBP		

Test Flow:

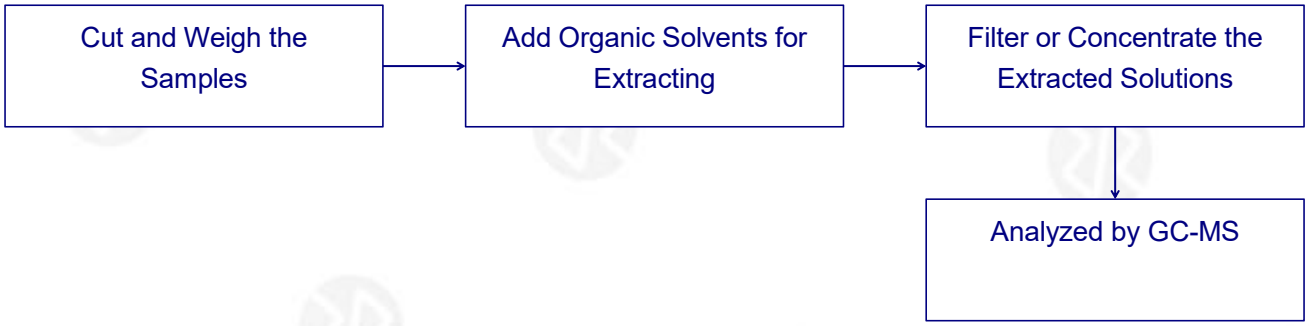
1 XRF Screening



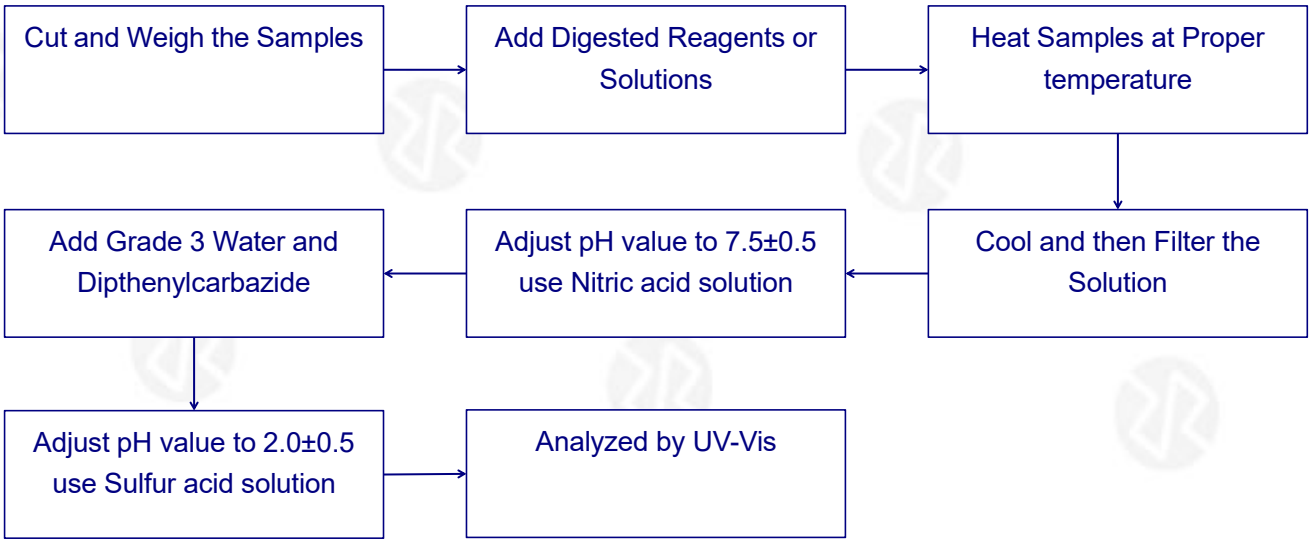
2 Lead, Cadmium, Mercury



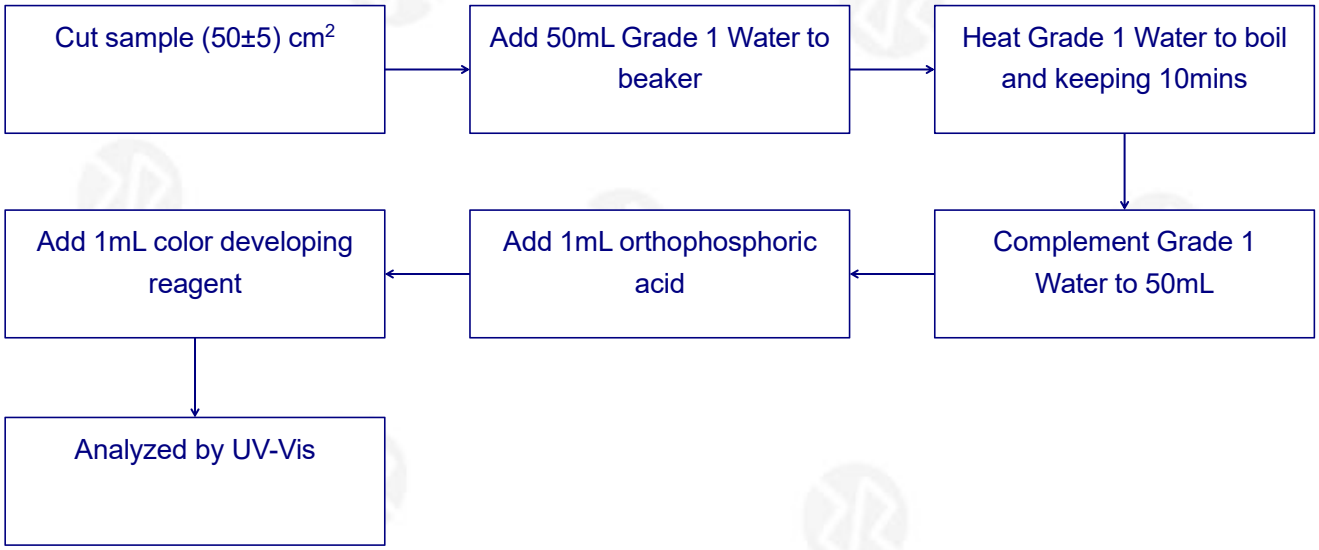
3 PBBs & PBDEs, Phthalates



4 Chromium(Cr VI) (Non-metal)



5 Chromium(Cr VI) (Metal)



Test Results:

Sample No.	Sample Description	Restricted Substances	XRF Testing	Chemical testing (mg/kg)	Conclusion
1	PCB	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
2	SMD Resistor	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
3	Silvery metal	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	NA	-	
		PBDEs	NA	-	
		DEHP	-	NA	
		BBP	-	NA	
		DBP	-	NA	
		DIBP	-	NA	

Sample No.	Sample Description	Restricted Substances	XRF Testing	Chemical testing (mg/kg)	Conclusion
4	SMD Capacitor	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
5	Chip bead	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
6	LED	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	

Sample No.	Sample Description	Restricted Substances	XRF Testing	Chemical testing (mg/kg)	Conclusion
7	Crystal Oscillator	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
8	IC	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
9	IC	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	

Sample No.	Sample Description	Restricted Substances	XRF Testing	Chemical testing (mg/kg)	Conclusion
10	Silvery metal	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	NA	-	
		PBDEs	NA	-	
		DEHP	-	NA	
		BBP	-	NA	
		DBP	-	NA	
		DIBP	-	NA	
11	IC	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
12	SMD Capacitor	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	

Sample No.	Sample Description	Restricted Substances	XRF Testing	Chemical testing (mg/kg)	Conclusion
13	Crystal Oscillator	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
14	SMD Resistor	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	
15	IC	Pb	BL	-	PASS
		Cd	BL	-	
		Hg	BL	-	
		Cr(VI)	BL	-	
		PBBs	BL	-	
		PBDEs	BL	-	
		DEHP	-	ND	
		BBP	-	ND	
		DBP	-	ND	
		DIBP	-	ND	

Notes:

1 Interpretation of screening results by X-ray fluorescence spectrometry (XRF)

1.1 XRF screening limits for different materials:According to IEC62321-3-1:2013.

Materials	Concentration (mg/kg)				
	Cd	Cr	Pb	Hg	Br
Metals	BL≤(70-3σ)<X<(130+3σ)≤OL	BL≤(700-3σ)<X	BL≤(700-3σ)<X<(1300+3σ)≤OL	BL≤(700-3σ)<X<(1300+3σ)≤OL	NA
Polymers	BL≤(70-3σ)<X<(130+3σ)≤OL	BL≤(700-3σ)<X	BL≤(700-3σ)<X<(1300+3σ)≤OL	BL≤(700-3σ)<X<(1300+3σ)≤OL	BL≤(300-3σ)<X
Composite material	BL≤(50-3σ)<X<(150+3σ)≤OL	BL≤(500-3σ)<X	BL≤(500-3σ)<X<(1500+3σ)≤OL	BL≤(500-3σ)<X<(1500+3σ)≤OL	BL≤(250-3σ)<X

(a) “OL” = “over limit”.

(b) “BL” = “below limit”.

(c) “X” = “Inconclusive”, result is intermediate between “OL” and “BL”.

(d) “NA” = “Not Applicable”.

1.2 For regulated substances[polybrominated biphenyls (PBBs), polybrominated diphenyl ethers (PBDEs)], the results showed the total Br content, while for regulated substance Cr(Ⅵ), the results showed the total Cr content. If the quantitative results for the elements Br and/or Cr are higher than the limit (for Br calculated based on the stoichiometry of Br in the most common congeners of PBB/PBDE), the sample is "inconclusive".

1.3 Where further investigation is necessary, and further chemical testing by ICP-OES (for Cd, Pb, Hg), UV-Vis (for Cr(VI)) and GC-MS (for PBB/PBDE) are recommended to be performed.

2 Interpretation of results by chemical tests.

2.1 mg/kg = milligram per kilogram = ppm = 0.0001%

2.2 “NA” = “Not Applicable”.

2.3 “ND” = “Not Detected”(<MDL).

2.4 “LOD” = Limit of Detection

2.5 MDL & LOD in wet chemical testing.

Item	Cr(VI)		Pb	Cd	Hg	PBB	PBDE	DEHP	BBP	DBP	DIBP
Units	μg/cm²	mg/kg									
MDL	0.10	10	10	5	10	10	10	50	50	50	50
LOD	1000		1000	100	1000	1000	1000	1000	1000	1000	1000

2.6 Boiling-water-extraction: According to IEC 62321-7-1:2015, determined of Cr(Ⅵ) on metal sample by boiling water extraction test method, and result is shown as Positive/Negative.

(a) Negative = Absence of Cr(VI) coating / surface layer: the detected concentration in boiling-water-extraction solution is less than 0.10μg with 1cm2 sample surface area.

(b) Positive = Presence of Cr(VI) coating / surface layer: the detected concentration in boiling-water-extraction solution is greater than 0.13μg with 1cm2 sample surface area.

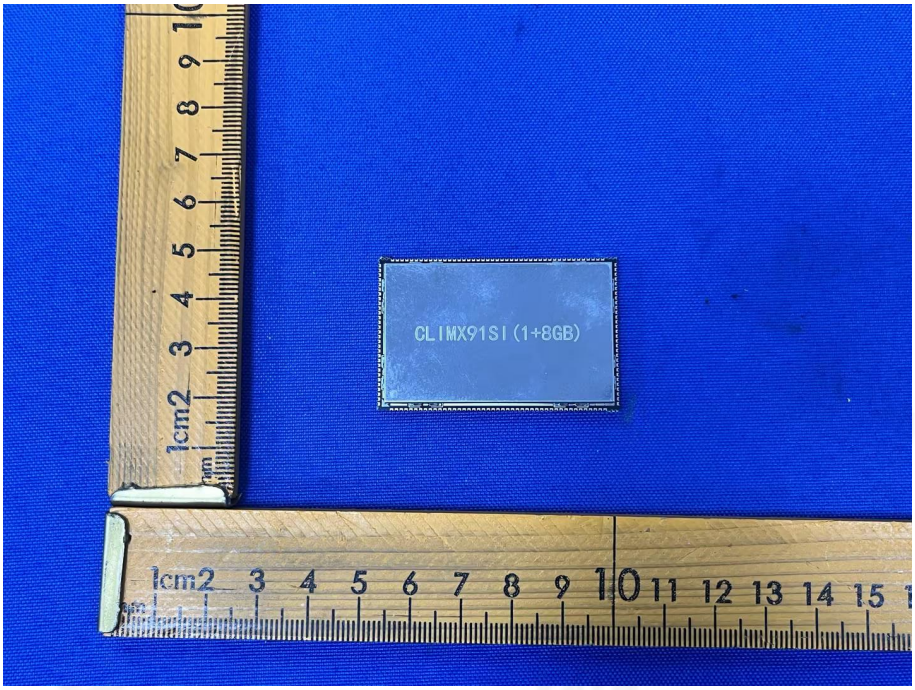
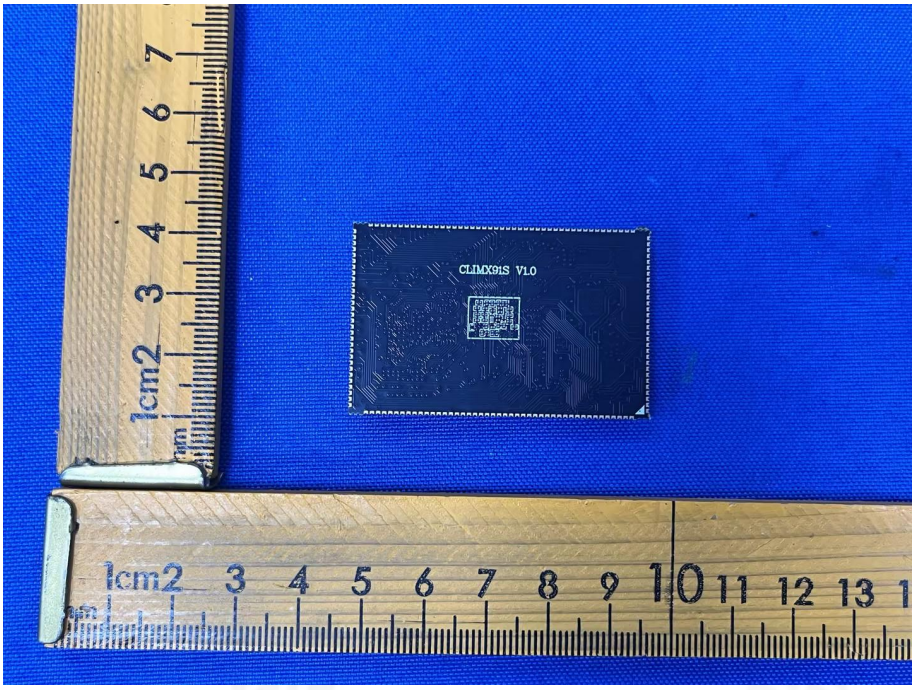
(c) Information on storage conditions and production date of the tested sample is unavailable and thus

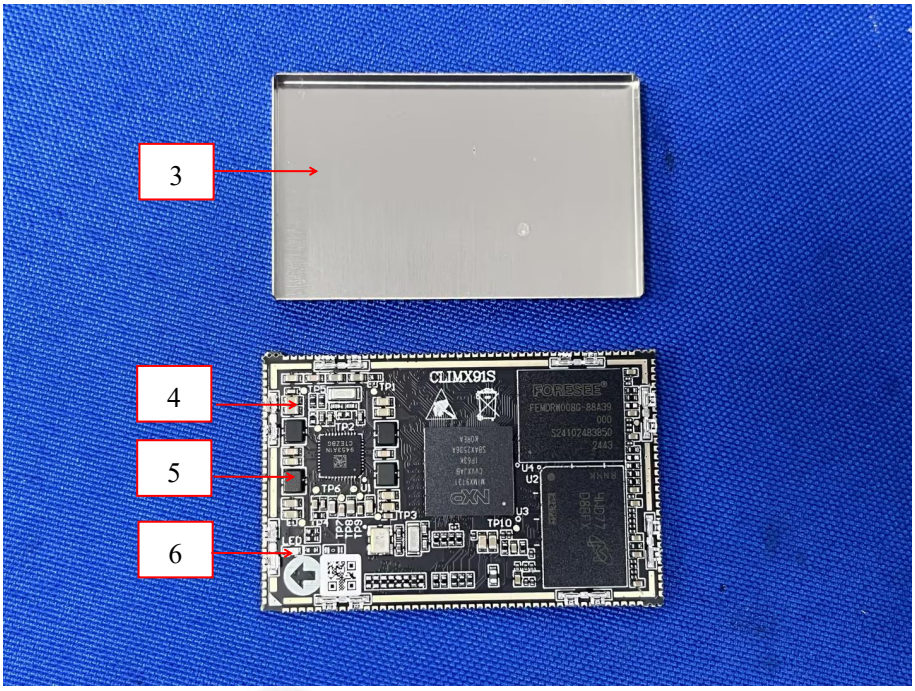
Cr(Ⅵ) results represent status of the sample at the time of testing.

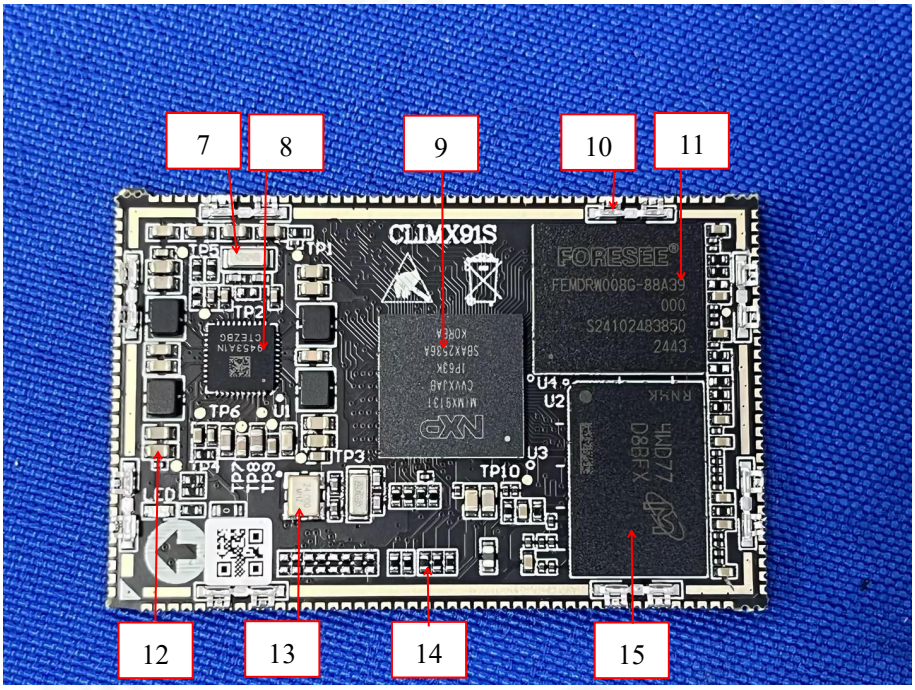
2.7 “DBP” = Dibutyl phthalate, “BBP” = Benzyl butyl phthalate, “DEHP” = Bis(2-ethylhexyl)-phthalate, “DIBP” = Diisobutyl phthalate .

3 As specified by client, only test the designated sample.

Submitted Sample Photo

1.	
2.	

3.	
4.	

5.	

※※※※※ **END OF REPORT**※※※※※

Statement

1. The information as listed on the first page of this test report was all provided by the client except the received date, testing period, test result(s) and test request. The client shall be responsible for the representativeness of sample and authenticity of materials, for which Shenzhen ZKT Testing Technology Co., Ltd. shall bear no responsibilities.

2. The test report is effective only with both signature and specialized stamp. Without written approval of Shenzhen ZKT Testing Technology Co., Ltd., this report can't be reproduced in full or in part.

3. This test data is only responsible for the tested sample. Shenzhen ZKT Testing Technology Co., Ltd. is not responsible for any economic and legal responsibility for the use of the test data, the direct or indirect losses resulting from the use of the test and all legal consequences.

4. If there is any objection to the report, client shall lodge a complaint with the company on the 30 working days from the date of receipt of the report. No late application will be accepted.